Final project: Analyzing How Artificial Intelligence Influences the Efficiency and Speed of Radiologists When Interpreting Radiographs

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1. Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
CXR	Chest X-ray
EMR	Electronic Medical Record
NLP	Natural Language Processing
CNN	Convolutional Neural Network
BoW	Bag of Words
TF-IDF	Term Frequency–Inverse Document Frequency
ROC AUC	Receiver Operating Characteristic – Area Under the Curve
KNIME	Konstanz Information Miner (Analytics Platform)
SVM	Support Vector Machine
NER	Named Entity Recognition
FDA	Food and Drug Administration (U.S.)
ICU	Intensive Care Unit
DICOM	Digital Imaging and Communications in Medicine
CT	Computed Tomography
MIMIC-CXR	Medical Information Mart for Intensive Care – Chest X-ray Dataset
CheXNet	A Deep Learning Model for Detecting Pneumonia from Chest X-rays
NegBio	Negation and Uncertainty Detection Tool for Radiology Reports
CheXpert	Chest X-ray Dataset with Uncertainty Labels
BERT	Bidirectional Encoder Representations from Transformers
Power BI	Business Analytics Tool by Microsoft
CPU	Central Processing Unit

2. Introduction

The Motivation behind our project is that we aim to help Radiologists who experience high workloads whereby they interpret hundreds of X-rays within a day, also to help the patients who might need immediate attention.

Chest X-rays are one of the most common and important imaging modalities in clinical medicine (Johnson et al.). They are quick, cheap, and provide fast assessment of the health of the lungs and heart and may be used in the evaluation of diseases including heart failure, lung tumors, pneumonia, and many other diseases. Although they are an important and widely utilized clinical diagnostic tool, traditional interpretation of chest X-rays has several major drawbacks. Interpretation requires training and expertise in order to accurately evaluate images. Even experienced radiologists may make small mistakes, or might miss small details, due to their heavy workload. This may slow down patient care. Radiologists have been in short supply world-wide. For example, a 2015 survey found there was one radiologist for every 12 million people in Rwanda (UK National Screening Committee). This has led to poor turnaround times of chest X-ray results. Delays in interpreting chest X-rays have become a significant safety concern in fast-paced health care systems like the UK's NHS and the U.S. Department of Veterans Affairs (UK National Screening Committee). These delays highlight the pressing need for tools to expedite and improve the accuracy of radiological assessments.

Recently, developments in artificial intelligence have exhibited incredible promise in addressing this concern. Benefitting from advances in computer vision and deep learning, researchers are now employing convolutional neural networks in medical imaging, and some remarkable outcomes have been generated (Rajpurkar et al.). A significant step forward was realized in 2017 when ChexNet, a deep learning model that detects pneumonia on chest X-ray images, was released—it reportedly performs at the same level as expert radiologists (Rajpurkar et al.). This moment captured considerable excitement about how AI might enhance the radiology workflow and potentially automate certain activities. In the time since 2017, the trend has continued, with increasing oversubscription for AI models that identify other medical conditions, such as lung nodules, fractures, and tuberculosis. Some systems are capable of "multi-label classification," which means the system identifies multiple conditions based on a single input image.

Artificial intelligence operates not only in identifying present conditions but is capable of inferring future ones (Rajpurkar et al.). For example, some recent research has demonstrated how an AI model analyzing a chest X-ray can predict a patient's likelihood of heart failure or even their survival time based solely on varied minutiae found in the image (Lotter et al.). This concept paves the way for novel screening possibilities. Consider a patient who has routine chest X-ray images taken in addition to their normal results; there may also be an indication that the patient has some elevated risk for a specific disease. Such information could result in earlier referrals, adjustments to lifestyle, or preventative therapy—all collected from data that was gained anyway (Rajpurkar et al.). In addition, the use of AI on chest X-rays with other clinical data, such as laboratory tests or symptoms of the patient, may also aid in predicting which patients may go on to worsen both in terms of trajectory of the disease or in clinical needs (Lotter et al.). This would provide a physician more data to act on sooner and with more confidence.

With expanding access to large annotated radiographic databases and advances in AI, deep learning algorithms (such as convolutional neural networks) are being developed to assist radiologists. It remains necessary to investigate whether AI actually enhances radiologists' interpretation time and accuracy and to what degree such AI aids diagnostic efficiency.

These remarkable developments have been facilitated by the creation of large datasets and advancements in algorithms, and multiple models are achieving performance levels same as or in certain instances, even surpassing the performance of human experts in controlled situations (Irvin et al.).

One of the striking developments is artificial intelligence's ability to find patterns even human vision often misses (Gichoya et al.). For example, few models can understand patients' characteristics like age, sex, and even race directly from the X-ray images—all these details cannot be seen by normal human vision (Gichoya et al.). Even though these capabilities are impressive, it raises questions about how AI models come to their decisions and whether the patterns they have found are clinically meaningful signals or were they being influenced by the hidden biases in the data (Lotter et al.).

To evaluate and train such models, researchers have turned to large, diverse datasets like MIMIC-CXR, which was released in 2019—this open database includes over 370,000 chest radiology reports (Johnson et al.). The data has a wide range of patient types, and most of them were ICU and inpatient cases, and even consists of scans from both portable and fixed X-ray machines, providing a valuable variety in both clinical and technical contexts. MIMIC-CXR is important because it addresses one of the biggest challenges which are being faced in medical artificial intelligence, particularly the need for a wide range of high-quality data (Johnson et al.). Before the introduction of artificial intelligence in the medical field, researchers often struggled due to limited datasets. In many ways, it plays a similar role for medical imaging as ImageNet has done for general computer vision research (Johnson et al.).

The dataset’s integration with other MIMIC databases like ICU vital signs and lab results has opened doors to a huge range of multi-model studies, where artificial intelligence models can learn from both images and clinical data (Johnson et al.).

A. Problem Statement

1. The primary challenges are:
   - Manual interpretation is slow, error-prone, and not scalable.
   - Radiologists always need help answering the question, "Is this report critical?"
2. The group proposed the following to address this:
   - Automated classification of radiological reports using machine learning
   - Sentiment analysis for clinical comprehension
   - Improving radiologists' decision-making accuracy and speed

3. Literature Review

1.MIMIC-CXR Dataset and Labeling

The MIMIC-CXR dataset has served as a fundamental resource in chest X-ray AI research established by Johnson and Huo et al. (Johnson et al.). This massive de-identified dataset includes thousands of chest radiographs with clinical radiology reports. It aims to provide the researchers with the means to train and test machine learning models in real healthcare settings. One of the most foundational innovations of this research is the way the images are being labeled. Since the original radiology reports were in unstructured narrative text, the team utilized natural language processing (NLP) to extract structured labels from the original reports (Peng et al.; Irvin et al.).

They specifically used two open-source NLP tools: NegBio, which identifies negations and uncertainties identified in narrative text (Peng et al.), and CheXpert, which classified findings as referring to either: present, absent, or uncertain (Irvin et al.). By combining both tools, every image could be tagged with one or more findings, such as Cardiomegaly, Consolidation, Pleural Effusion, and No Finding. One detail noted in the narrative text that both tools allowed the researchers to capture is if the two tools disagreed in their diagnosis of any of the aforementioned, the researchers tagged the result with a Disagreement Flag.

In sum, the result was a multi-label dataset that encompasses a broad range of conditions, from the very common to the quite rare. To assist with the accuracy of benchmarking, a subset of 2,000 images was also manually reviewed and labeled based on the expert opinion of an experienced radiologist (Johnson et al.). In addition, the authors provided predefined train, validation, and test splits to help minimize variability in evaluating models across studies. One of the strongest features of the dataset is that it is realistic. Labels are derived from actual clinical reports so therefore they exhibit the way radiologists communicate, even including some imprecision. However, this has also produced some issues. For example, if a result was not mentioned in a report, then the absence of a finding was therefore considered negative by default even though it simply was not addressed in the report (Johnson et al.). There is also the problem of class imbalance, as the No Finding label appears in approximately one-third of cases, while positive findings such as Fracture or Pleural Other are very rare.

2.AI Tools in Chest Radiography

In addition to datasets from research, there has been an increasing number of AI tools developed for clinical chest X-ray analysis. One well-known example is the Siemens Healthineers' AI-Rad Companion, which is cleared by the FDA and uses deep learning to help identify five different findings such as pulmonary lesions, pleural effusion, consolidation, and atelectasis. The software highlights suspicious areas of the image, allowing radiologists to view these indications in addition to their examination. In an evaluation completed in 2023, researchers tested the AI tool with a total of 499 chest X-rays against radiologist interpretations, supported by CT scans in some cases (Niehoff et al.). The AI system discovered lung lesions at an 83% sensitivity, which is better than the radiologist's 52% in the original report, consolidation at 88% compared to the radiologist's 78%, and lung collapse at a sensitivity of 54% compared to the original 43% sensitivity of the report (Niehoff et al.). Importantly, there is a tradeoff with the AI system, as there are more false positives, and in some cases, the artificial intelligence report indicated something was present when it was not, decreasing precision. However, AI-Rad Companion can still be useful because it has a very high negative predictive value, meaning that when it said there wasn't anything there, it was usually correct. This is particularly valuable as it can be used as a safety net, allowing radiologists to confirm their decisions on normal cases and fix missed diagnoses (Niehoff et al.).

Research competitions and startups such as Lunit and Aidoc have also contributed to the field, creating deep learning algorithms that perform at a radiologist level and have perfect accuracy when designed to do so for certain tasks.

3.Cross-Institutional Model Performance

One frequent concern that appears in research is the difficulty of getting artificial intelligence models to work well in hospitals. Earlier research had models trained using one dataset, such as the NIH ChestX-ray14 (Wang et al.) or CheXpert (Irvin et al.), and reported acceptable performance. But when those same models were tested on data from different hospitals, the performance readings would drop. A good illustration is the Zech study published in 2018. They trained a model that could detect pneumonia with chest X-rays collected from only one hospital and reached a reasonably high accuracy threshold with an AUC of approximately 0.93 (Zech et al.). When the model was tested on chest X-rays from a different hospital dataset, its accuracy dropped to around 0.81 (Zech et al.). Additionally, even when they trained using data from multiple hospitals, the issue did not completely dissipate. The model had learned to take shortcuts such as concluding if an image was collected from a portable X-ray machine, which coincidentally aligned with visiting patients with pneumonia more frequently. The model became so adept with these features that it could even tell which hospital collected the X-ray with greater than 99% accuracy (Zech et al.).

In a more recent publication, Fernández-Miranda in 2024 embarked on a similar investigation. They found that a COVID-19 model was accurate with images from the same brand of X-ray machine it was trained on, but did not perform as well with images from different brands of X-ray machines (Fernández-Miranda et al.). Even slight differences in the way the image processing was performed led to significant differences in the artificial intelligence model's reliability. All this to say, just because an artificial intelligence model is reliable in one place, it does not mean it can be generalized anywhere (Fernández-Miranda et al.). This is why researchers are starting to pay more attention to external validation and domain adaptation methods, which ensure that models are reliable in the context of different hospitals or differing equipment. Another component to validating generalizability is the research around federated learning. This is a process that allows hospitals to work collaboratively in training artificial intelligence models with data from multiple locations but does not require sharing patient data (Fernández-Miranda et al.). Again, this may represent a step forward towards developing artificial intelligence methods that are both accurate and generally applicable.

4. Gaps in Current Research

Artificial intelligence has advanced tremendously in analysing chest X-rays, yet there are still a number of real-world challenges to tackle. Below are some of the largest gaps still present in the present content:

1. Models Don't Always Work Well for Everyone, Everywhere

The main issue that arises is that artificial intelligence models show promising results in the lab, but do not translate effectively to the real world. One model's flair for X-ray interpretation learned from one machine does not always transfer to another brand or model. Subtle differences in contrast, sharpness, noise, etc., can confuse the artificial intelligence model (Fernández-Miranda et al.). It is not just the machines. The models may also behave differently depending on the patient's age, sex, or race. Some models even have been found to infer race from chest X-rays, something that no human radiologist can do (Gichoya et al.). Of course, this raises questions of implicit bias—whether the model could in fact be differentiating one patient from another without any human practice. There are instances when scholars research the information for different groups, but it is nothing that is regularly done. We need to evaluate these factors earlier on in the process and use better training to ensure equity for everyone.

2. Doctors Need to Know the "Why," Not Just the "What"

If an AI, for instance, tells a doctor that there is a 90% chance of pneumonia without a justification, most doctors would not be comfortable accepting this. Right now, there are many AI systems that are black boxes (Kallianos et al.). They provide an answer but do not show the underlying reasoning. Some systems support generalization by providing heatmaps to indicate highlighted areas in the image that the model was supposedly looking at, but heatmaps are frequently too variable (Das et al.). What doctors want is a plain explanation that resembles human reasoning. That's the type of language they use when talking amongst themselves, and they want a format that they can discuss with each other. We need better explanations. We also want to begin testing AI in realistic environments of practice so that we can better understand at what point, if any, different explanations lead to changes in trust in AI results.

3. What Works in One Hospital Might Not Work in Another.

Just because an AI model has been shown to perform well within a specific hospital doesn't mean it performs just as well when utilized in a different environment. Hospitals are different from each other in all manners—the patients they treat, the equipment they have in the hospital, to the types of conditions they will most frequently see. If a hospital sees a lot of cancer patients, for example, many of their chest X-rays might have chemotherapy ports placed within them. The AI might see those ports associated with a certain set of diseases and then make errors when it was performed on a patient in a hospital where those devices are rare (Zech et al.). This is a major concern.

One potential way to remedy this is to train models on data from multiple hospitals, and ideally from different regions and countries. In this way, the AI gets some sense of what chest X-rays look like outside of a specific instance (Lin et al.). However, work on datasets like this is still being developed. In the meantime, research is looking to understand how to support models to potentially adapt to the new environments they will be used in while minimizing their errors and training on the new case.

4. Rare But Serious Conditions Get Missed

Most datasets of chest X-ray images will include common diagnoses such as pleural effusion or 'no finding', while diseases that are both severe and rare diagnoses premiums, such as pneumothorax, will only be represented much less frequently in a dataset (Wang et al.). This is problematic for AI models because they often learn to unify what they see the majority of the time, leading to the potential for omittance of severe but rare presentations altogether (Das et al.). Some researchers are employing various strategies to deal with rare presentations of disease, such as over-sampling the cases of diseases that are uncommon, or changing the training so that it attends to extreme cases better, but these strategies are largely yet to be validated (Das et al.). Adding to this problem, many studies are simply returning an average accuracy score over all findings; averaging can disguise the model's potential false negatives when it misses a finding that is more undesirable (Zech et al.). In conclusion, we need more research demonstrating that AI tools not only perform well as an average model, but also when it counts, specifically to identify overlay of life-threatening but rare conditions.

5. Synthetic Data Sounds Cool — But It's Tricky

To compensate for the difficulty of obtaining rare cases, some researchers are developing AI technology that creates synthetic X-rays (Chen et al.). The synthetic images serve to balance training data and teach the model how to recognize conditions it might not see very often otherwise. Synthetic images can have some downsides too. If those images are too like real images, then perhaps the model just memorizes the images and gives an inflated sense of the model's performance (Chen et al.). On the other hand, if the synthetic images have weird quirks to them or just don't really look realistic, then the model would just pick off the wrong cues (Chen et al.). Then there are trust issues. If doctors know an AI application was trained on synthetic images, would they trust an AI tool's decisions for real patients? The answer is not clear yet (Eltorai et al.).

5. Methodological Framework

A. Knime Platform

We used KNIME, a visual analytics platform, to handle and organize the data (Berthold et al.). KNIME's intuitive drag and drop method was particularly helpful in building a modular low code pipeline that could be reproducible and easily shared across technical and clinical teams. The aim was to build a pipeline that could be shared transparently and quickly modified in collaboration with both programmers and non-programmers alike.

B. Model Selection

To address the critical issue of automating and enhancing the interpretation of chest X-ray (CXR) results, the team put in place a methodical machine learning pipeline. This pipeline, which was implemented using KNIME and Python, included the following essential steps: evaluation, model selection and training, feature extraction, and text preprocessing.

Text preprocessing was the first step, which is crucial when working with raw clinical notes. Sometimes clinical reports are messy and use a lot of different punctuation, words, and acronyms. Because of this, all of the words were lowercase and the text lacked punctuation and special characters. Once tokenization had separated phrases into words, stop-word removal was used to eliminate common, uninformative terms like "the" and "is." These procedures reduced noise in the ensuing analysis and prepared the data for numerical translation.

When converting the numeric forms that could be comprehended by machine learning algorithms, pre-processing was done to the text data. Two popular text vectorization techniques, Bag of Words (BoW) and TF-IDF (Term Frequency-Inverse Document Frequency), were employed. BoW generates a simple frequency table based on the presence of each word in each publication. They are separate from each other with no regard to location or context. It provides a very simple representation that is inexpensive to compute. TF-IDF, on the other hand, gives every word a weight depending on how important it is across the corpus. because long stories in clinical documents tend to hide a very important vocabulary.

Machine Learning Models: Several machine learning models were explored for classification, with a focus on distinguishing between important and non-critical reports.

1. logistic regression

Logistic regression is a linear model for binary classification problems. It predicts the probability that a given input belongs to a specific class using a logistic function. It was used as a baseline in this study due to its ease of use, interpretability, and speedy training on high-dimensional, sparse data generated by BoW and TF-IDF techniques. However, its performance was limited compared to more complex models.

2. The Random Forest

The Random Forest, a machine learning ensemble method, builds many decision trees and averages their output to boost the prediction and lower the overfitting. The model fitted best for the clinical report data because it can perform on high-dimensional features, is immune to overfitting, and can identify non-linear relationships. It performed best among other models with respect to ROC AUC (0.848), which indicates that it possesses an outstanding power of classification in differentiating critical from non-critical reports. Its strength and interpretability made it a good choice in the healthcare industry, where model decisions must be comprehensible.

3. XGBoost, Extreme Gradient Boosting

XGBoost is a high-speed, high-accuracy gradient-boosted decision tree algorithm widely used for its ability to provide accurate results. With its ability to iteratively improve model performance by minimizing classification error at each step, it is especially useful on complex data with intricate patterns.By regularization, XGBoost provided even more control over the bias-variance trade-off in this project to produce outcomes almost as good as the Random Forest ROC-AUC of 0.847. Therefore, even with an imbalanced and small dataset, it proved useful in maximizing generalizability.

C. Data Handling with KNIME

We started with a review of MIMIC-CXR, which includes not only metadata with patient IDs, study IDs, and image paths, but also structured labels derived from the CheXpert and NegBio tools, and, where needed, free-text radiology reports (Johnson et al.; Irvin et al.; Peng et al.). KNIME's file reader nodes facilitated the upload of the typically large CSV files, which were often compressed. Given the MIMIC-CXR size, it was important to address duplicated records.

The KNIME Duplicate Row Filter allowed us to determine and eliminate duplicate records ultimately every patient would be unique (Berthold et al.). Additionally, while any inconsistencies at the label level were addressed by dataset curators, we needed to verify that every instance of label data matched the corresponding report.

Normalization was instrumental in aligning the values of our data across different dimensions. For the use of text data from reports, we isolated and sanitize the relevant portions of the reports such as the Findings and Impression. For label normalization, we conscientiously decided to treat the uncertain values as negative, which we justified as we followed a conservative and accepted practice (Irvin et al.). This was all done using KNIME's Rule Engine nodes to maintain reliability, completeness, and transparency throughout the evaluation process.

The MIMIC-CXR dataset contains timestamped studies, enabling us to incorporate a temporal review into our dataset. We utilized Group by and Sort nodes to arrange each patient's studies in temporal order. An index to denote the order of studies, and a structured label trajectory over time (Johnson et al.). These features that we engineered proved important for any temporal or progression-based modelling where we kept the order of studies in mind. Though mostly complete, there were some records that had missing data. For studies that did not have reports, and therefore did not have labels, we opted to remove these studies.

KNIME was a fantastic environment for this whole preprocessing workflow (Berthold et al.). Its visual, node-based system made each step auditable and allowed our clinical collaborators to check outputs without having to read code. The workflows could be annotated and exported, which made sharing easy. KNIME also supports database integration and Python/R scripting, which means you have the freedom to scale or customize steps whenever it is needed (Berthold et al.). This tool was most useful in allowing us to prototype a variety of preprocessing paths filtering low quality images, filtering per study types, or adjusting how labels were mapped—all without having to write entirely new scripts. At the conclusion of this phase, we had a clean and organized data set: images for modelling, labels format for supervised learning, and additional, time-aware features for deeper analyses.

D. Descriptive Analytics and Exploratory Visualization

Prior to starting the model training process, we wanted to dive into the dataset (Johnson et al.). This provided us the opportunity to examine the patterns, verify our preprocessing steps, and shape important downstream modeling decisions. We began with a distribution of clinical labels across the MIMIC-CXR dataset. As anticipated, "No Finding" was the most frequently observed label, appearing in approximately one-third of studies (Johnson et al.). Next, Support Devices, which is largely a reflection of the ICU context in which many patients have tubes, catheters, or other medical devices visible on the X-ray, appeared in approximately 29 percent of studies.

Among the next most common pathological conditions, Pleural Effusion appeared in about 23 percent of cases, Lung Opacity in 22 percent, and Atelectasis appeared in 20 percent (Johnson et al.). The "rare" conditions revealed Lung Lesion in around 2.7 percent of cases, Fracture in 1.7 percent, and Pleural Other in less than one percent. The distribution of the labels clearly indicated class imbalance would need to be taken seriously when training models (Irvin et al.). In order to better understand and communicate the differences, we created bar charts that provided both our technical and clinical contributors with a snapshot of label prevalence.

We also looked at the kinds of X-ray views in the dataset. About a third of the images were frontal views (AP or PA) and about two-thirds were lateral views (Johnson et al.). These proportions are typical in hospital imaging, particularly in the intensive care unit; portable X-ray machines are often used, and patients tend to be restricted to a single frontal image (probably AP).

While the dataset does not specify which X-ray machines were utilized, we sampled DICOM metadata and confirmed the use of at least three different brands of imaging equipment (Johnson et al.). Due to privacy and data use agreements, we do not confirm any brands, but we were relieved to see some differences in imaging sources. The differences are beneficial when we train models that generalize to data from other hospitals or imaging devices. Nevertheless, we acknowledge that image quality and style may differ based on the machine used to capture the image hence we decided to move forward with the text reports rather than the images.

An even distribution of time suggested time-series modeling may be an interesting path to pursue, particularly as a way to monitor changes in conditions over the hospitalization (Johnson et al.). We also investigated what clinical findings often tended to co-occur. We tended to see patterns present themselves immediately using correlation heatmaps (Johnson et al.). As an example, Edema, Pleural Effusion, and Enlarged Cardio mediastinum tended to appear together, which makes sense in the context of heart failure. Consolidation and Airspace Opacity were often found together, which aligns with the use of the terms in practice. Pneumothorax co-occurred with Support Devices, which likely indicated that many of these patients had chest tubes (Johnson et al.). These relationships were of course interesting in a medical domain but also provided us with useful clues for our heads-up displays of structured model outputs. A system that could learn to reflect labels that demonstrate their relationship to one another in a multi-label or hierarchical approach may be even more accurate and clinically relevant (Irvin et al.).

In concluding the exploratory stage, we conducted a handful of manual spot checks, where we viewed individual X-rays with their radiology reports and assigned labels (Johnson et al.). For example, one report stated that the right side of the chest was opacified, noted pleural fluid, and stated suspicion for hydropneumothorax: the labels Pleural Effusion, Pneumothorax, and Opacity were exact matches to that report. Based on these reports we were able to determine the sentiments, whether it is a positive sentiment or negative , what level of severity it has as well .

E. Using and Understanding Radiology Reports (Jing et al.; Shahin et al.)

Important aspect of our work pertains to the "free text" radiology reports. These reports are highly informative and go beyond simply listing findings; they also specify the location of abnormalities, how certain they are, and how things are changing over time (Jing et al.). We wanted to see how this detailed text could be incorporated into the model, including exploring simple summary creation (Shahin et al.).

Research interest has also been increasing in the idea of automatically generating the entire report from images, similar to what an AI assistant could use for drafting a radiology note (Shahin et al.). That's beyond our current project scope, though, and we decided to do a much simpler version, like just generating the "Impression" section that is typically a shorter summary (Jing et al.).

This activity is much closer to image captioning, and prior work has demonstrated success through combining a CNN with an attention-based language model, where they reasonably work for images with text outputs (Jing et al.). We also manually reviewed a subset of results to ensure clinical accuracy and to ensure that the generated texts were not only linguistically similar but also clinically correct based on evidence-based medicine (Shahin et al.).

6. Model Performance and Results

1. The models were validated using ROC curves, confusion matrices, and accuracy measures. The ROC AUC metric provided a performance measure of the model for all thresholds of categorization, allowing comparison between true positive and false positive rates. This was especially useful in the medical realm, where reducing false negatives, missing a valuable case is of utmost importance. Together, they ensured that the choice of the model was based on careful measures of performance. The Rain forest and XGBoost performed the best.

2. The data utilized in this endeavor stems from a repository containing a cumulative size of 89,587 medical records of patients, where each row variables denoting the noted presence and absence of a variety of radiological conditions (Johnson et al.). Such conditions may include, but are not exclusive to, pleural effusion, edema, collapse, pneumothorax, and other conditions (Irvin et al.). Each record will also contain a pertinent sentiment label whether a positive or negative label indicative of the clinical tone of the report derived from the radiologist's notes (Peng et al.). Additionally, basic demographic information will be collected and an indicator specifying whether the patient has noted medical history. As the data is organized in this predetermined format, we can leverage and utilize the data to model and create dashboards, which allows the potential for exploring patterns across a large patient population (Johnson et al.).
3. Clinical tone of a report is an essential aspect of the analysis and is the classification of sentiment, which signifies if the clinical tone of a radiology report suggests a normal or concerning finding (Peng et al.). In this dataset, negative sentiment is much more commonplace, with 61,444 records indicating negative sentiment, or roughly 69% of records. Positive sentiment has appeared in 28,143 records, or about 31% of

records. While this imbalance may be seen in clinical practice, it is not surprising in the clinical environment, particularly with diagnostic imaging, where we are often asked to consider patients with specific symptoms, or a confirmed concern (Irvin et al.). This analysis is well represented in the dashboard shown in the bar chart and indicates negative findings.

4. Overall, clinically this may indicate that the patients received routine diagnostics or have higher caseloads of emergency or diagnostic imaging rather than the typical check-up role (Das et al.).

5. The demographic data show a considerable imbalance between genders in the dataset. Namely, of the patients in the dataset, 60.5 percent were male (54,215), while slightly fewer were female (39.5% or 35,372) (Johnson et al.). Compiling the data graphically in pie or bar chart form often conveys rapid visual performance of the available gender representation in the data. There may be many reasons for the large gender imbalance seen in the dataset. Perhaps it is a reflection of the patient flow in a specific clinical situation or site from which the data originated, it demonstrates trends that are either male- or female-based related to the conditions such as pneumothorax or edema seen in this patient population, or it is simply analogous to the demographic characteristics of the population sampled (Gichoya et al.). No matter the reason given, it is important to recognize gender imbalance in analytical representation of a dataset when constructing machine learning models. Failing to account for gender variation in the data, especially if the sample is heavily imbalanced between genders, could prejudice the objectivity of the algorithms gained from the data model (Gichoya et al.).

6. The dataset provides useful information about the different radiological conditions described in the reports. Most frequently, pleural effusion is documented, with a total of 38,357 cases (Irvin et al.). Collapse is seen in 23,096 cases. Edema is noted in 22,033 reports, with pneumothorax in 22,491 cases. Tube placement usually signifies some sort of invasive treatment, such as the placement of a chest tube, and is documented in 17,370 cases. Consolidation has been cited 9,052 times. All of these frequency findings are presented on a bar graph located on the dashboard so that clinicians can quickly see the findings they observe most frequently (Johnson et al.). This information can be very helpful. A high incidence of pleural effusions may suggest that pneumonia or heart failure was seen frequently. The respiratory index may indicate a high volume of emergency or post-op care if there has been labeled a high frequency of tube placements. These patterns can help to inform hospital administrators and clinical researchers to allocate resources where they are most needed, or where areas of study may be relevant (Kallianos et al.).

7. The dashboard serves to visualize information but the backend uses KNIME for predictive modeling (Berthold et al.). The KNIME workflow processes files by reading them, handling missing values, and filtering columns for data preparation.

The workflow continues through various trainings of machine learning models including Random Forest, Support Vector Machine, and Gradient Boosted Trees (Berthold et al.). The models predict outcomes such as report sentiment or the risk of studying particular conditions based on patient characteristics. At the end of the workflow, results are measured using metrics such as ROC curves and scoring nodes (Berthold et al.). While the performance metrics don't get reported into Power BI, they can be sent back into the dashboard for visual comparison and to allow users to understand historical and predictive information.

8. One of the most impressive features of the Power BI dashboard is its drill-down interactivity, leading users from broad observations to highly specific clinical subsets (Das et al.).

7. Conclusion

This study demonstrated the effective use of machine learning techniques in the automation of sentiment analysis and report classification of chest X-ray radiology. By careful preprocessing, feature engineering, and model experimentation, we established a pipeline with the capability to classify critical reports with high accuracy.
We performed sentiment analysis based on rule-based thresholds to provide clinical sentiment scores that enable escalation of serious and infectious cases.
Both Random Forest and XGBoost, with ROC AUC values of 0.848 and 0.847, respectively, outperformed the models in processing complex, unstructured medical text data. The application of these models by the visual dashboard again showed the value of the system in helping radiologists through streamlining report triage and accelerating decision speed. As a whole, the project shows the way AI can be an effective aide in radiology to develop data-driven, scalable health solutions.

8. Future Directions and Research Questions

AI in chest X-ray analysis is moving quickly, and while our current workflow is a strong foundation, there remains much exciting ground to cover (Rajpurkar et al.; Wang et al.). Based on what we've done so far and the challenges we've observed, here are some key areas for future research focus.

1. The application of BERT based Natural Language Processing (NLP) models in radiology has revolutionized the task of radiologists interpreting and reporting imaging exams. BERT is a contextual language comprehension deep learning model that can be trained on large amounts of radiology reports to identify clinical vocabulary, detect major findings, and normalize language across the reports. This facilitates real-time feedback while typing or dictation, automated report generation, and even intelligent differential diagnosis recommendations from the findings identified. Furthermore, as a further quality control feature, these models can assist in

detecting report errors or omissions. Apart from speeding up reporting, BERT-based NLP improves overall accuracy, consistency, and readability of clinician-to-clinician communication by assisting radiologists with such time-consuming but important tasks.

2. An agile, web-based application for clinicians should be implemented in order to enhance the capability of AI in radiology. Such an application would allow physicians to input observations either through structured forms, free text, or voice dictation and, concurrently, obtain AI-supported feedback based on NLP and image analysis algorithms. Real-time interaction may include auto-reporting completion, suggested templates based on scan types, and even real-time summaries that highlight critical abnormalities. Visual markers on radiographs, for example, bounding boxes or heatmaps, can be useful for clinicians to rapidly confirm AI-flagged issues. Moreover, when this interface is integrated with EMR systems, relevant patient history and laboratory results may be viewed in conjunction with imaging studies, further enhancing clinicians' workflow. A well-designed platform would also allow for collaboration by allowing multiple users to see and comment on the same case, and could include learning features that allow junior radiologists to learn through comparing new cases to comparable previous cases.

3. An important next step is to assess how well our model performs on chest X-rays outside of MIMIC (Johnson et al.). While it is impressive to know that the model performs well on training data, it is another achievement for it to perform equally well on images from a different hospital (Zech et al.). External validation is critical. To develop truly generalizable AI, we must assess other datasets like CheXpert, NIH ChestX-ray14, or partner with a local hospital (Irvin et al.). If our model underperforms on external data, adjustments are necessary.

4. Techniques like domain adaptation (adjusting a model learned from one dataset for application to another) and meta-learning (training models to be robust across varying datasets) can help (Lotter et al.). A future where institutions share a benchmark for collaborative model evaluation. Where one institution trains and others test could advance generalizability research (Lin et al.).
5. federated learning: Instead of moving data between hospitals which is difficult due to privacy concerns federated learning trains models locally on each hospital’s server, exchanging only model updates to enable global learning across institutions (Fernández-Miranda et al.). Yet, questions arise: How do we prevent one institution's data from dominating learning? How do we handle significant variations across institutional datasets? (Fernández-Miranda et al.)

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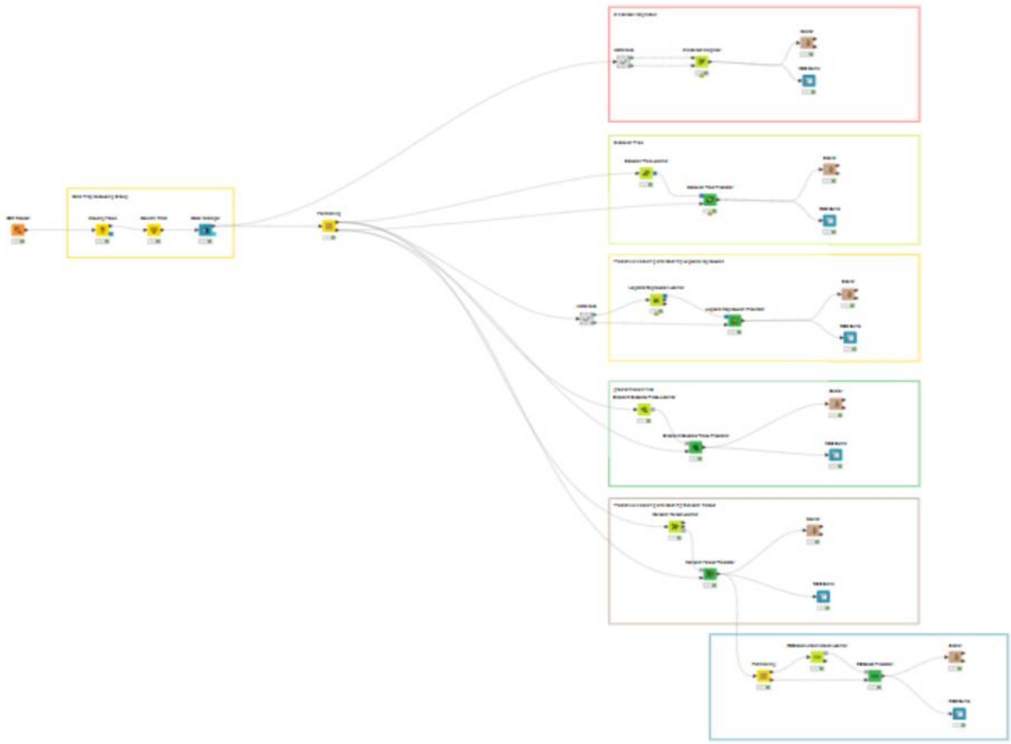
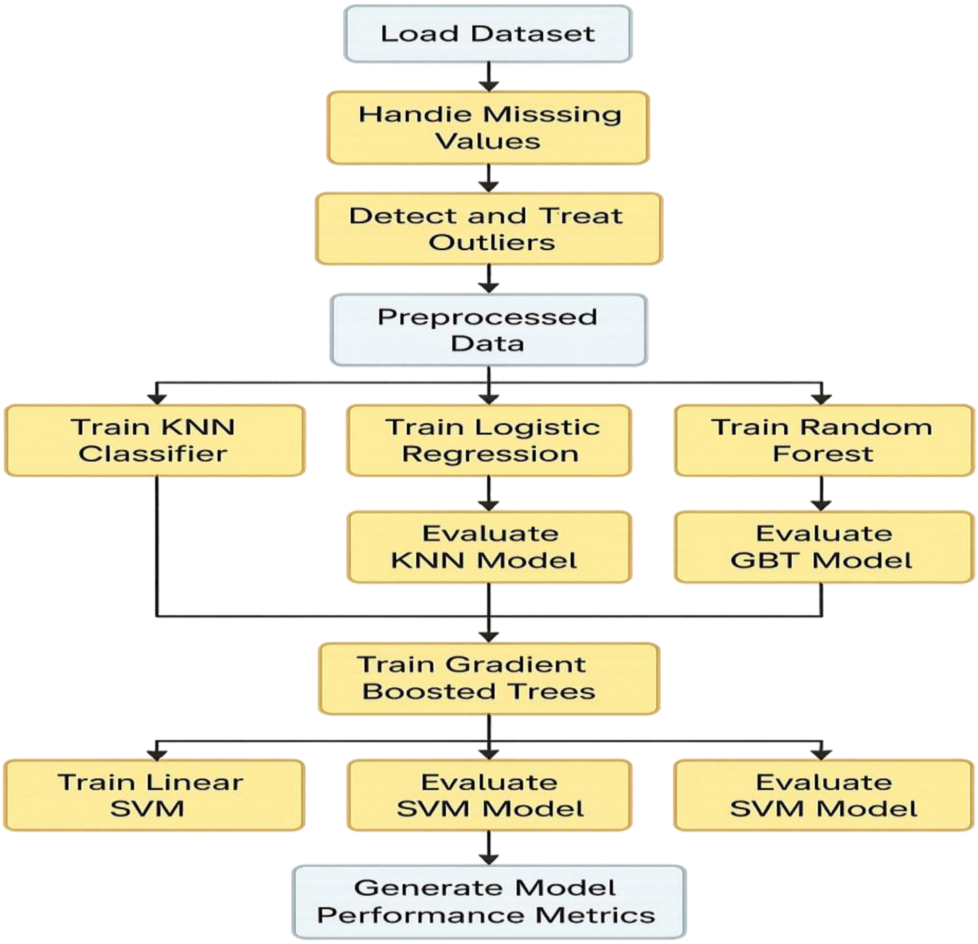
Appendix / Appendices

1. Knime Workflow
   https://hub.knime.com/amanjhagta/spaces/Public/MIMIC-CXR~6WM-

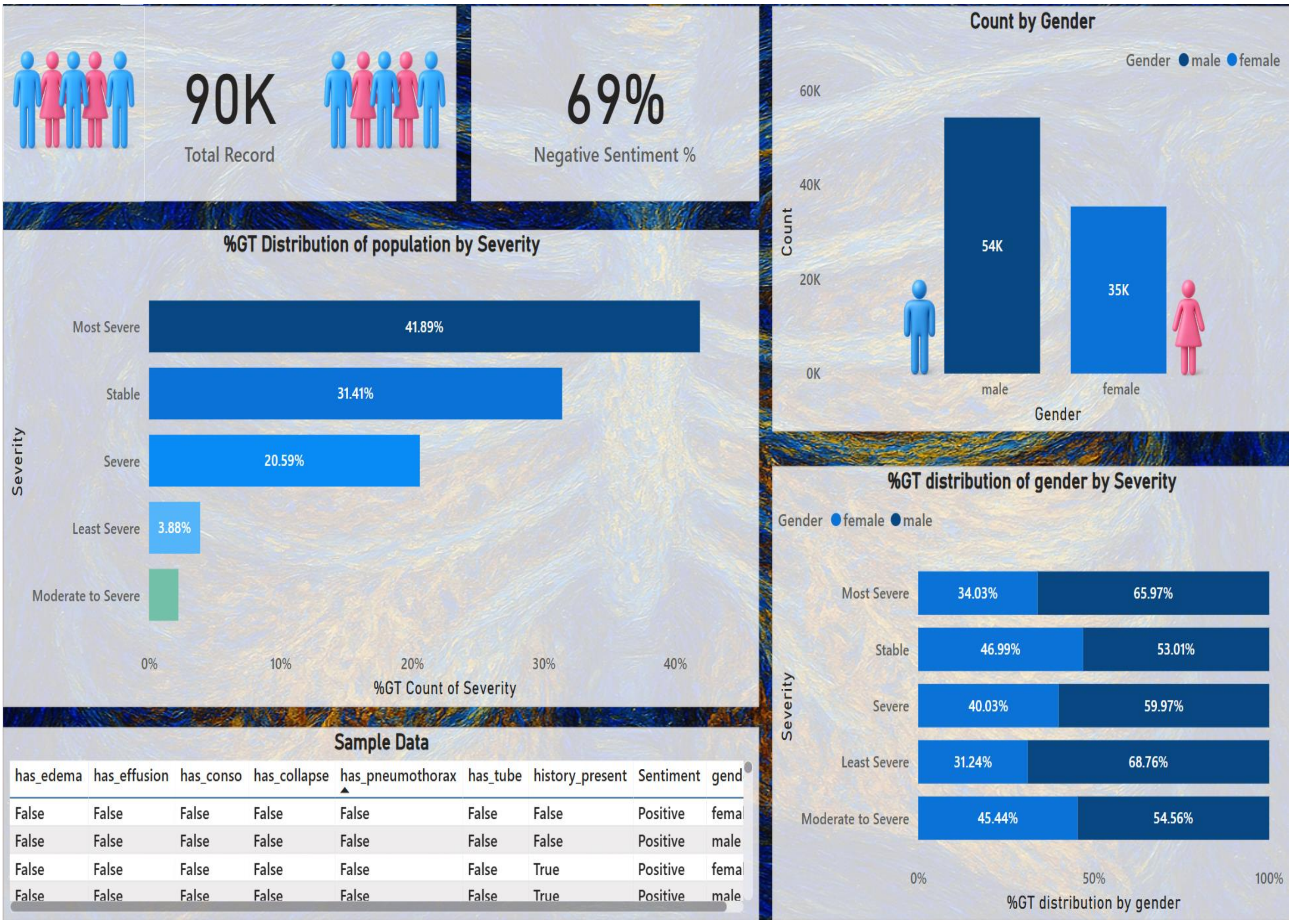


23U9yf5eOogA/current-state

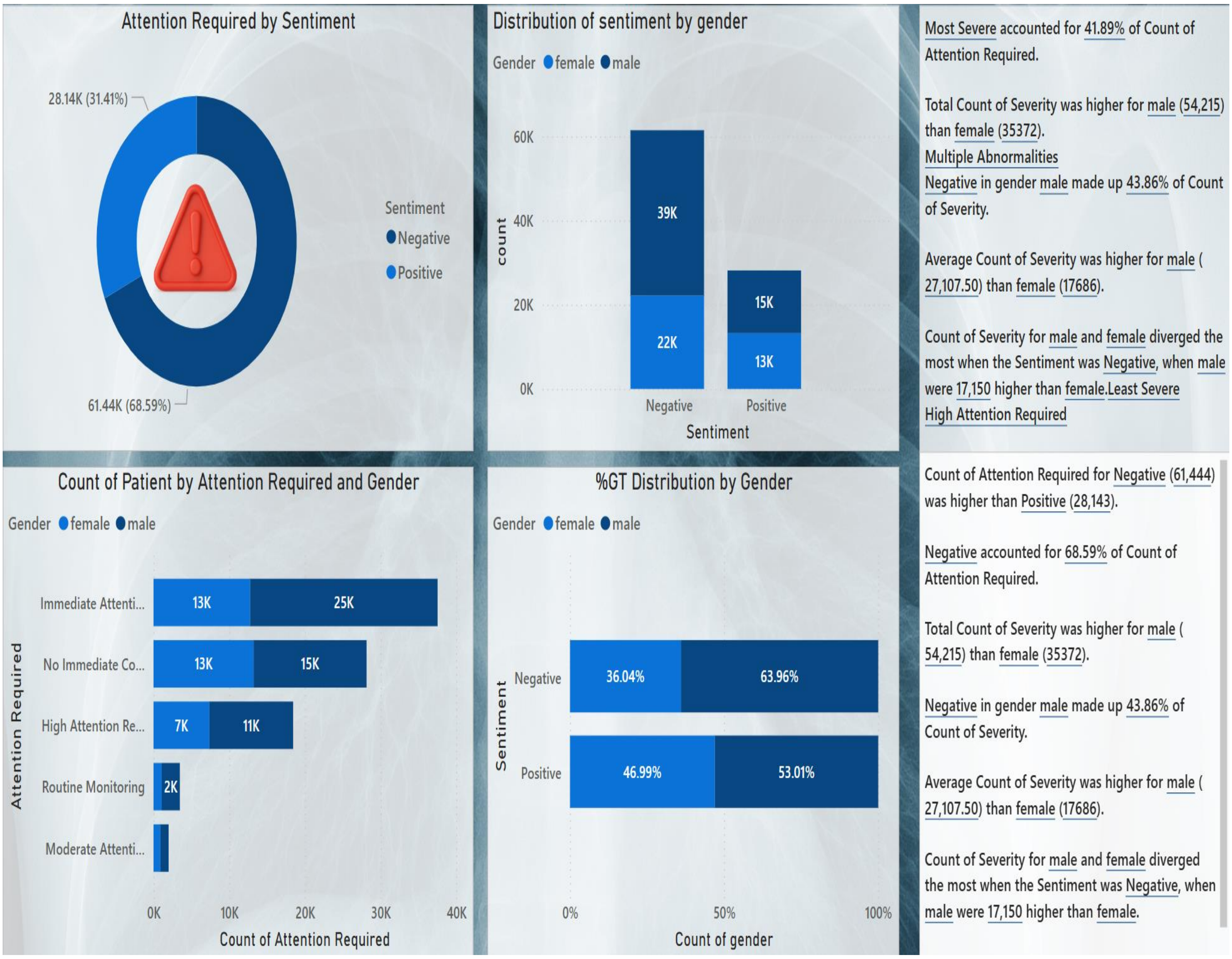
A Diagram Of Our Workflow :



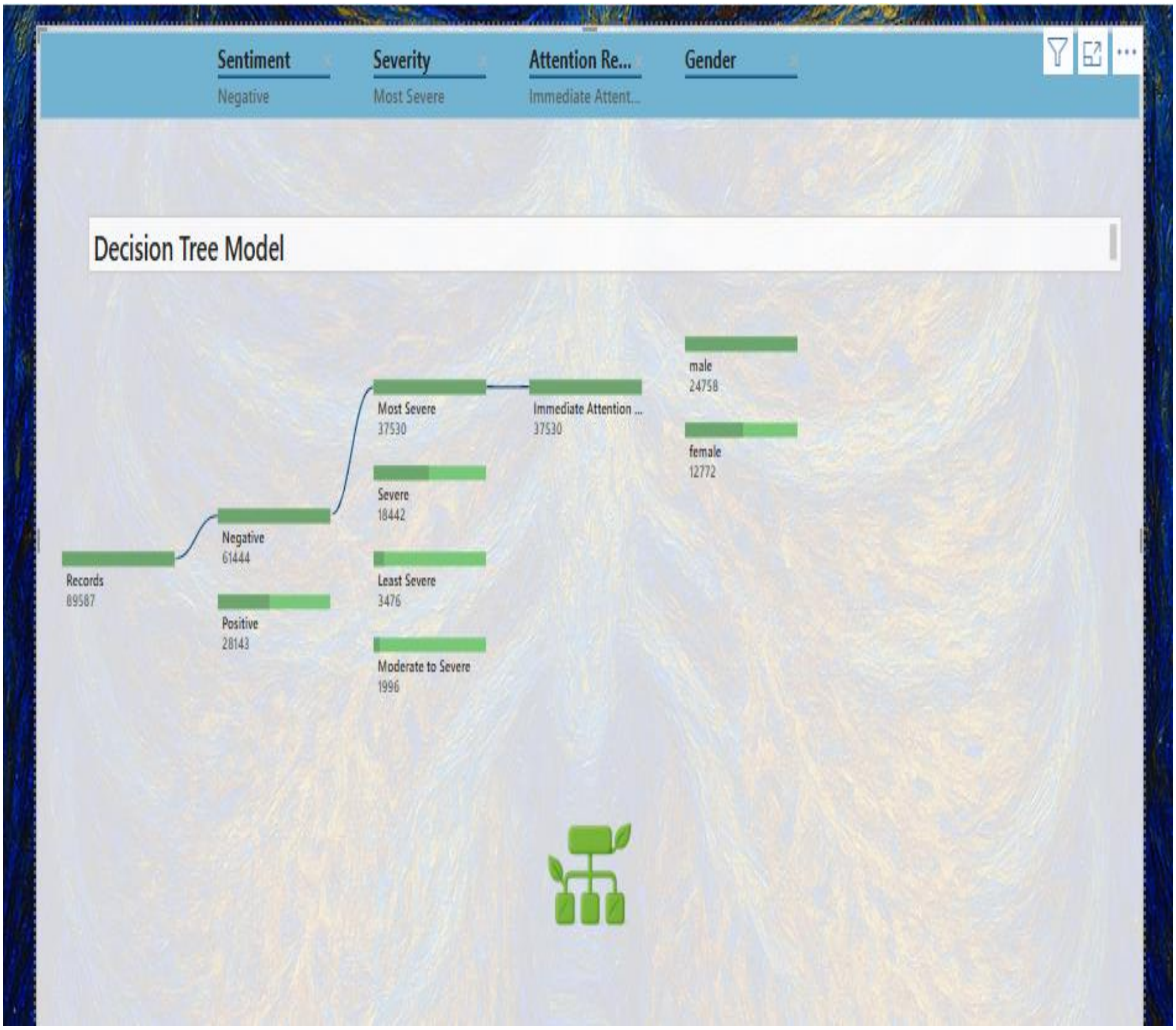
Dashboards:-



90K
Total Record
69%
Negative Sentiment %
Count by Gender
Gender male female
60K
40K
20K
0K
Count
54K
35K
male
female
Gender
%GT Distribution of population by Severity
Most Severe
Stable
Severe
Least Severe
Moderate to Severe
Severity
41.89%
31.41%
20.59%
3.88%
0%
10%
20%
30%
40%
%GT Count of Severity
%GT distribution of gender by Severity
Gender female male
34.03%
65.97%
46.99%
53.01%
40.03%
59.97%
31.24%
68.76%
45.44%
54.56%
0%
50%
100%
%GT distribution by gender
Sample Data
has_edema has_effusion has_conso has_collapse has_pneumothorax has_tube history_present Sentiment gend
False False False False False False False Positive femal
False False False False False False False Positive male
False False False False False False True Positive femal
False False False False False False True Positive male

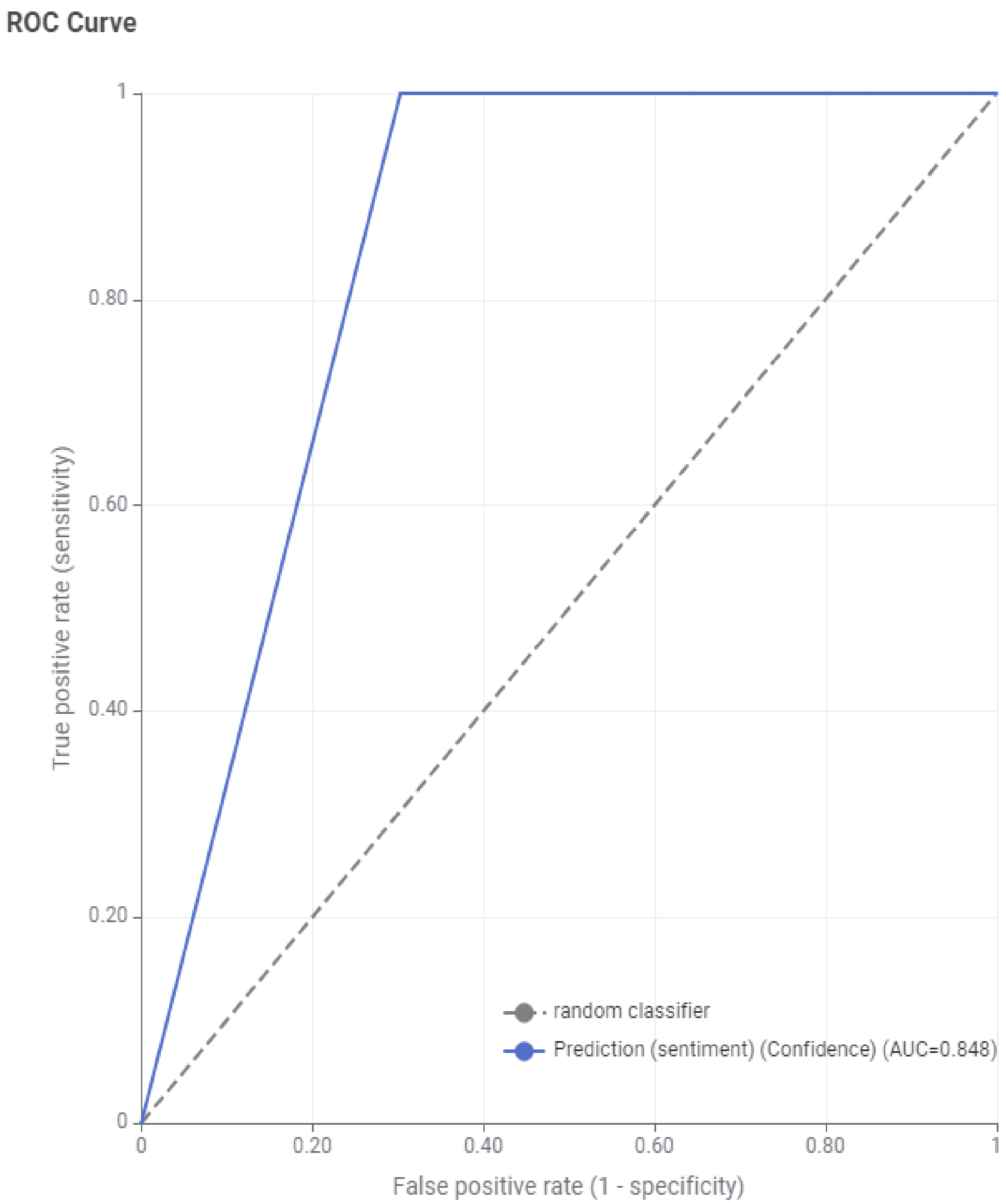


Attention Required by Sentiment
28.14K (31.41%)
61.44K (68.59%)
Sentiment
Negative
Positive
Distribution of sentiment by gender
Gender female male
60K
40K
20K
0K
count
39K
22K
15K
13K
Negative
Positive
Sentiment
Most Severe accounted for 41.89% of Count of Attention Required.
Total Count of Severity was higher for male (54,215) than female (35372).
Multiple Abnormalities
Negative in gender male made up 43.86% of Count of Severity.
Average Count of Severity was higher for male (27,107.50) than female (17686).
Count of Severity for male and female diverged the most when the Sentiment was Negative, when male were 17,150 higher than female.Least Severe
High Attention Required
Count of Patient by Attention Required and Gender
Gender female male
Immediate Attenti...
No Immediate Co...
High Attention Re...
Routine Monitoring
Moderate Attenti...
13K
25K
13K
15K
7K
11K
2K
Attention Required
0K
10K
20K
30K
40K
Count of Attention Required
%GT Distribution by Gender
Gender female male
Sentiment
Negative
Positive
36.04%
63.96%
46.99%
53.01%
0%
50%
100%
Count of gender
Count of Attention Required for Negative (61,444) was higher than Positive (28,143).
Negative accounted for 68.59% of Count of Attention Required.
Total Count of Severity was higher for male (54,215) than female (35372).
Negative in gender male made up 43.86% of Count of Severity.
Average Count of Severity was higher for male (27,107.50) than female (17686).
Count of Severity for male and female diverged the most when the Sentiment was Negative, when male were 17,150 higher than female.

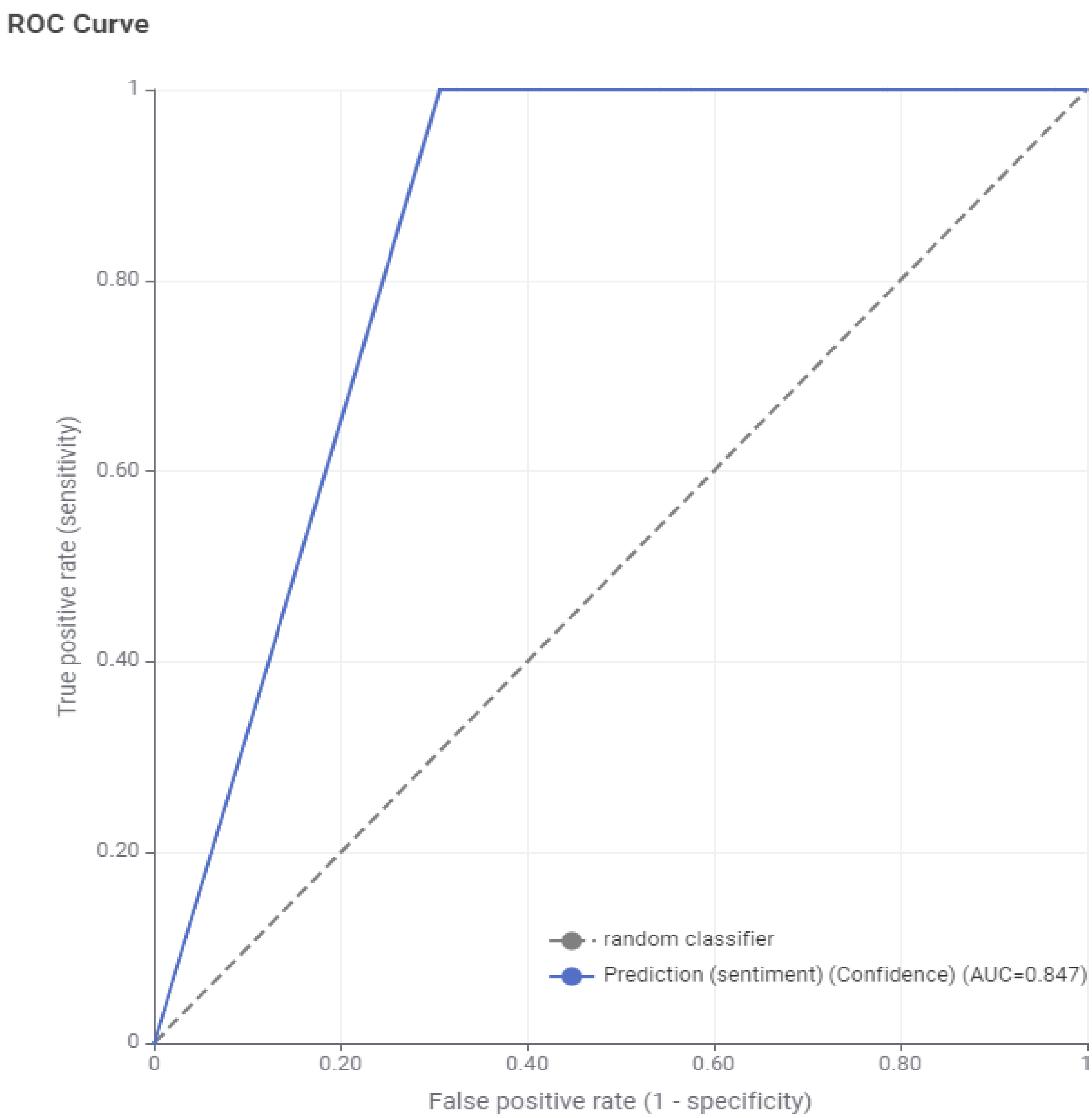

Sentiment
Negative
Severity
Most Severe
Attention Re...
Immediate Attent...
Gender
Decision Tree Model
Records
89587
Negative
61444
Positive
28143
Most Severe
37530
Severe
18442
Least Severe
3476
Moderate to Severe
1996
Immediate Attention ...
37530
male
24758
female
12772

1. Results
   Random Forest: **0.848**
   XGBoost: **0.847**

ROC CURVE – RANDOM FOREST



ROC CURVE – XGBOOST



2. **Data Processing on Python**

Github: https://github.com/Amanjhagta/Data-Science/blob/main/Scripts/MIMIC-CXR.ipynb

Code:

1. **Importing main libraries**

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

2. Extracting the data from files

```
import os
import pandas as pd
import re

folder_path = r"/content/drive/MyDrive/TDSP/New folder"  # Update your path

# Store extracted data
all_data = []

# Define expected section headers to avoid false column splits
expected_headers = {"EXAMINATION","HISTORY", "COMPARISON",
"FINDINGS", "IMPRESSION", "TECHNIQUE"}
```

```python
# Recursively search for .txt files in all subfolders
for root, dirs, files in os.walk(folder_path):
    for file_name in files:
        if file_name.endswith(".txt"):  # Process only .txt files
            file_path = os.path.join(root, file_name)

            with open(file_path, 'r', encoding='utf-8') as file:
                lines = file.readlines()

            # Process data
            structured_data = {}
            current_key = None

            for line in lines:
                line = line.strip()
                if not line:
                    continue  # Skip empty lines

                # Match first occurrence of `:` but only if the key is in expected
headers
                match = re.match(r"([^:]+):\s*(.*)", line)

                if match:
                    key = match.group(1).strip()
                    value = match.group(2).strip()
```

```
            # Only process valid headers, otherwise append to previous key
            if key in expected_headers:
                structured_data[key] = value
                current_key = key  # Set current key for multiline values
            else:
                if current_key:  # Append to previous section
                    structured_data[current_key] += " " + line.strip()
        elif current_key:  # Handle multiline values
            structured_data[current_key] += " " + line.strip()

    # Add filename and folder path as identifiers
    structured_data["Filename"] = file_name
    structured_data["Folder_Path"] = root  # Store folder where file was
found

    # Append structured data to list
    all_data.append(structured_data)

# Convert list of dictionaries to a DataFrame
df1 = pd.DataFrame(all_data)

# Ensure all required columns exist (even if missing in some reports)
for col in expected_headers:
    if col not in df1.columns:
        df1[col] = None  # Fill missing columns with None
```

```python
# Display DataFrame
print(df1.head())
```

3. Importing biobert libraies

```python
from transformers import AutoTokenizer,
AutoModelForTokenClassification
from transformers import pipeline

# Use a pre-trained model that is fine-tuned for medical NER (for
example, ClinicalBERT)
model_name = "dmis-lab/biobert-large-cased-v1.1"  # Ensure it's the
right version fine-tuned for NER
tokenizer = AutoTokenizer.from_pretrained(model_name)
model =
AutoModelForTokenClassification.from_pretrained(model_name)

# Initialize a pipeline for Named Entity Recognition (NER)
ner_pipeline = pipeline("ner", model=model, tokenizer=tokenizer)

# Example text: You can replace this with your medical text
text = """
The patient was diagnosed with diabetes and hypertension.
He was prescribed Metformin and Lisinopril for treatment.
"""

# Use the NER pipeline to extract medical entities
entities = ner_pipeline(text)

# Print the extracted entities
for entity in entities:
   print(f"Entity: {entity['word']}, Label: {entity['entity']}, Score:
{entity['score']}")
```

4. Extracting the medical features from the dataset using the impression column

```
import pandas as pd
from transformers import AutoTokenizer,
AutoModelForTokenClassification, pipeline

# Publicly available general NER model
model_name = "dslim/bert-base-NER"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model =
AutoModelForTokenClassification.from_pretrained(model_name)

# Initialize the NER pipeline
ner_pipeline = pipeline("ner", model=model, tokenizer=tokenizer,
aggregation_strategy="simple")

# Define medical conditions and associated keywords
conditions = {
    "has_conso": ["consolidation"],
    "has_edema": ["edema"],
    "has_effusion": ["effusion"],
    "has_tube": ["tube", "intubation"],
    "has_collapse": ["collapse", "atelectasis"],
    "has_pneumothorax": ["pneumothorax"]
}

# NER-based condition detection function
def detect_conditions_with_ner(impression_text):
```

```
    if pd.isna(impression_text) or impression_text.strip() == "":
        return pd.Series({column: False for column in conditions})

    result = {column: False for column in conditions}

    try:
        entities = ner_pipeline(impression_text)
    except Exception as e:
        print(f"Error processing text: {impression_text}. Error: {e}")
        return pd.Series(result)

    extracted_terms = [entity['word'].lower() for entity in entities]

    for column, keywords in conditions.items():
        result[column] = any(keyword in extracted_terms for keyword in
keywords)

    return pd.Series(result)

# Example DataFrame


# Apply detection
df_conditions = df1["IMPRESSION"].apply(detect_conditions_with_ner)

# Merge and display results
df1 = pd.concat([df1, df_conditions], axis=1)
print(df1.head())
```

5. Extracting the gender from data

```
import pandas as pd
import numpy as np

# Sample DataFrame
```

```
# df = pd.read_csv("your_file.csv")  # if you're reading from a file
# Assuming the DataFrame has 'EXAMINATION' and 'HISTORY'
columns

def extract_gender(text):
    if pd.isnull(text):
        return None
    text = str(text).lower()
    if 'f' in text and any(w in text for w in [' f ','__f ','___f ', '_f ', ' f ', '___f
','female','___F','WOMAN','woman']):
        return 'female'
    if 'm' in text and any(w in text for w in [' m ','__m ','___m','___m ',' m '
'_m ', ' m ', '____m ','___M',' male','MAN', 'man ']):
        return 'male'
    return None

def determine_gender(row):
    # Check the 'EXAMINATION' column first
    gender = extract_gender(row['EXAMINATION'])
    if gender:
        return gender

    # Check the 'FINDINGS' column
    gender = extract_gender(row['FINDINGS'])
    if gender:
        return gender

    # Check the 'IMPRESSION' column
    gender = extract_gender(row['IMPRESSION'])
    if gender:
        return gender

    # If no gender is found, return a default message
    return 'could not find the gender'


def history_flag(history_text):
```

```python
    return bool(pd.notnull(history_text) and str(history_text).strip())

# Assuming df is your DataFrame
df1['gender'] = df1.apply(determine_gender, axis=1)
df1['history_present'] = df1['HISTORY'].apply(history_flag)
```

6. Removing unnecessary columsn

```python
df1.dropna(subset=['IMPRESSION'], inplace=True)

df1 = df1[df1['gender'] != 'could not find the gender']
```

7. **Getting the sentiments from the files**

```python
df1['Sentiment'] = df_conditions.any(axis=1).apply(lambda x: 'Negative'
if x else 'Positive')
```